Methodology for Urdu Replication Study

Extending the CHI 2026 Study on Deaf and Hard-of-Hearing Access to Intelligent Personal Assistants

Group Members

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1. Study Objective

The proposed study extends the original research to a Pakistani DHH audience by replicating its experimental design in Urdu. The extension introduces a new language, script, and local cultural context while retaining the core study structure so that the findings can be meaningfully compared with the original study.

2. Proposed Methodology

Step	Method
1. Instrument Translation & Validation	Translate the SUS, Adjective Scale, and NPS into Urdu, followed by back-translation and review with a Pakistani Sign Language (PSL) consultant. Validate the Urdu instruments before the main study.
2. LLM Touch Localization	Re-author the LLM prompts and task vocabulary for Urdu and a Pakistani smart-home context, while preserving the overall interaction structure of the original study.
3. Participant Recruitment	Recruit Pakistani Deaf and Hard-of-Hearing participants through relevant DHH schools and associations.
4. Experimental Design	Use the same three-condition, within-subjects design: Natural Deaf Speech, Facilitated English/WoZ-style speech adapted to the local study, and LLM-Assisted Touch.
5. Measures & Interviews	Collect SUS, Adjective Scale, NPS, and WER where applicable, together with semi-structured interviews conducted in Urdu or PSL.
6. Comparison	Compare the Pakistani/Urdu findings with the original study's findings, while interpreting differences in light of language and cultural context.

3. Why This Methodology Is Appropriate

The methodology preserves the original study's core experimental structure while adapting language, tasks, recruitment, and context for Pakistani DHH participants. This provides a practical basis for examining whether the original findings extend to an Urdu-speaking environment without changing the central research design.

4. Key Consideration

Translation should not be treated as validation by itself. The Urdu versions of the measurement instruments should undergo an appropriate validation process before they are used for the main experiment.

Source Basis

This methodology is based on the methodology proposed in the supplied presentation, which specifies translation/back-translation, LLM prompt localization, Pakistani DHH recruitment, the same three-condition within-subjects design, and comparison with the original study.